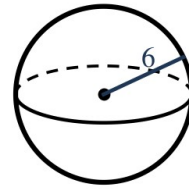


Speak Like A Mathematician:

sphere / great circle:

hemisphere:



Sphere Formulas

Surface area: $SA = 4\pi$ _____ .

Volume: $V = (4/3) \pi$ _____ .

Hemisphere: half the sphere's volume; total surface = half the sphere's surface **PLUS** the flat _____ .

Example 1 - Sphere with $r = 6$

- Find the surface area (exact and nearest tenth).
- Find the volume.

Example 2 - From the Diameter

A sphere has diameter 10.

- a) Find its surface area.
- b) Find its volume.

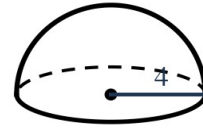
Example 3 - Working Backward

- a) A sphere has surface area 324π . Find its radius.
- b) Now find its volume.

Example 4 - Hemisphere

The hemisphere at the right has radius 4.

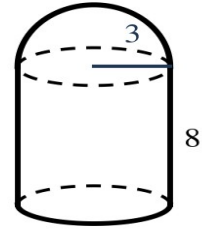
- a) Find its volume.
- b) Find its TOTAL surface area (dome plus flat face).



Example 5 - Composite Solid (Silo)

The silo at the right is a cylinder of radius 3 and height 8 with a hemisphere on top.

- Find the cylinder's volume.
- Find the hemisphere's volume.
- Find the total volume (exact and nearest tenth).



Example 6 - Scaling a Sphere

- A sphere's radius doubles. What happens to its surface area? Its volume?
- Why does a basketball hold much more than twice the air of a ball half its radius?